Fundamentals of MPRA Experiments

ISMB-ECCB 2025: *Tutorial IP2 (MPRAs and IGVF Resource)*

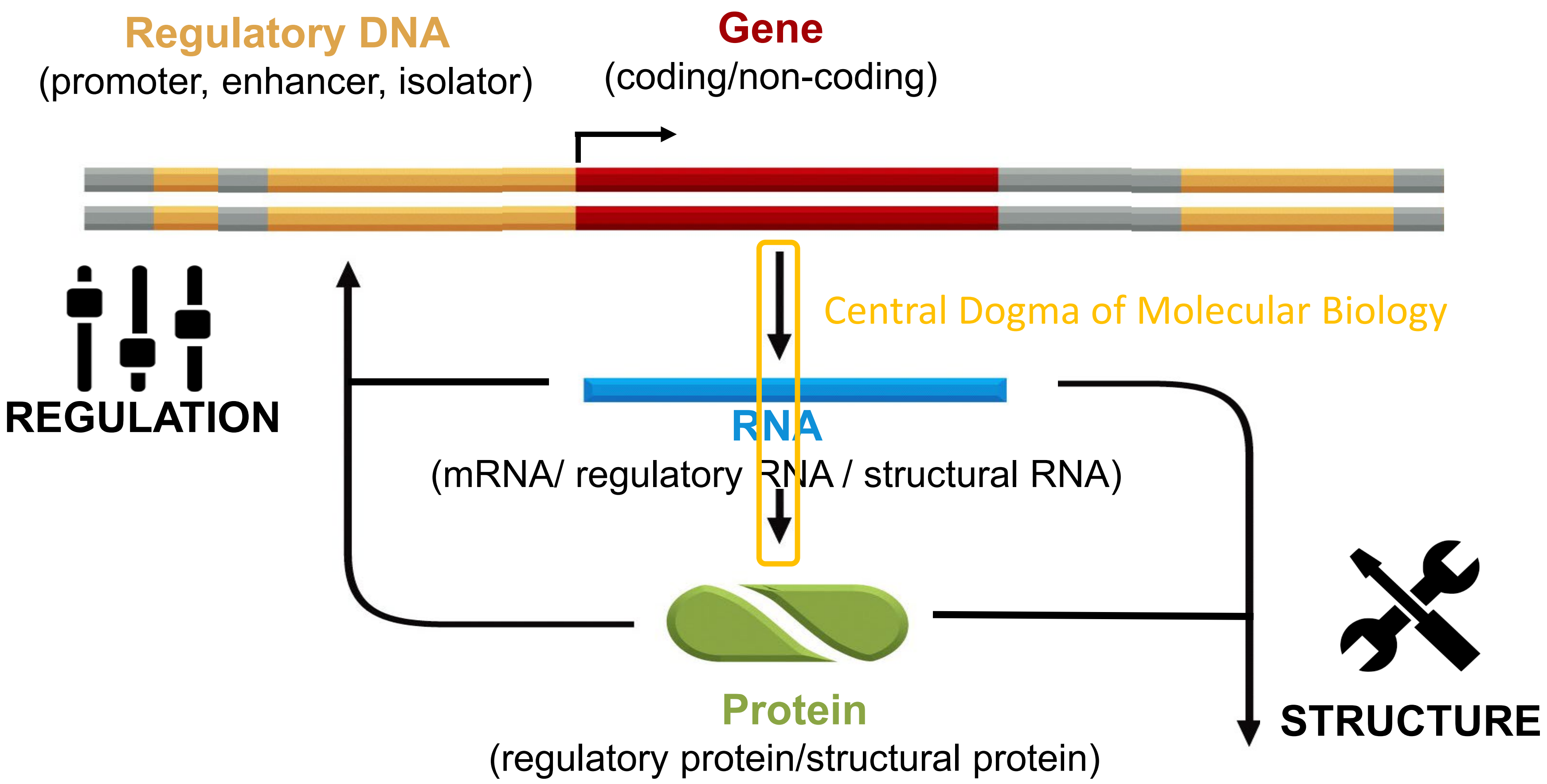
Martin Kircher
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025

UNIVERSITÄT ZU LÜBECK
UNIVERSITÄTSKLINIKUM Schleswig-Holstein
Berlin Institute of Health @Charité

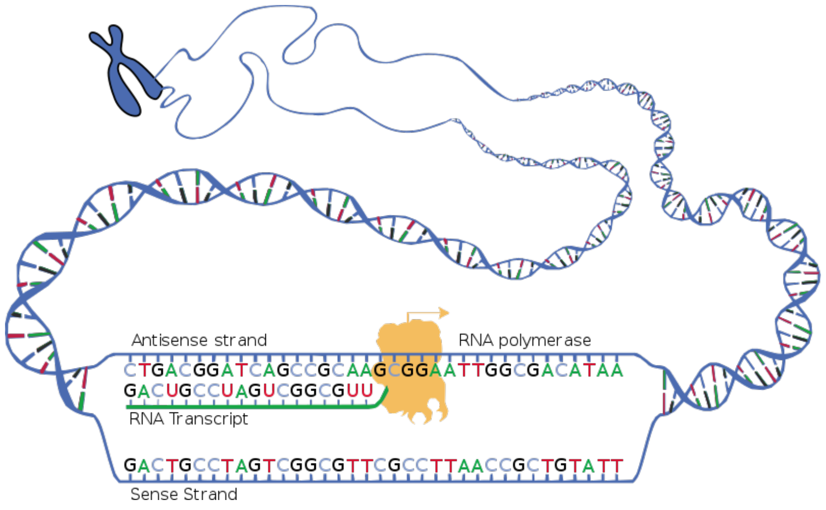
Part I – The underlying biology

Variant-to-function relation

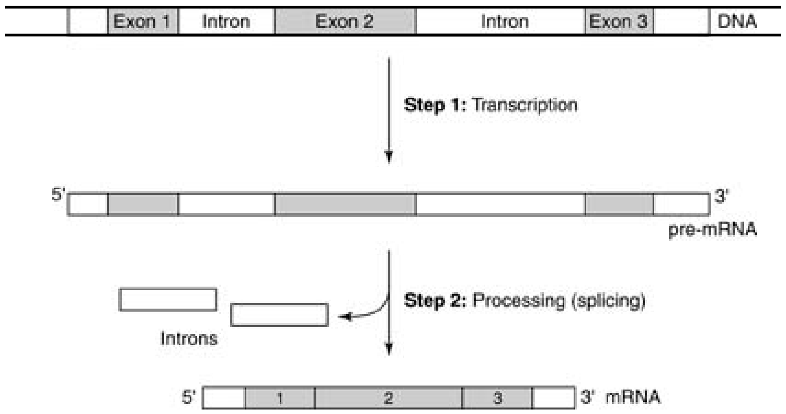


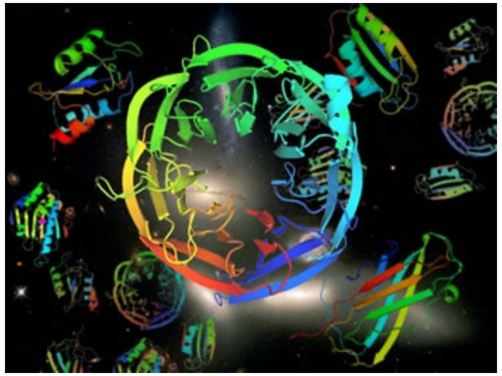
Complexity reminder...

- One gene many transcripts
- Exons/intron gene structures in eukaryotes
- Various steps of RNA processing
- RNA interference
- Cell-type specificity
- Protein modifications and complexes
- Protein protein interactions, catalytic activities, metabolite concentrations, …
- …



https://www.genome.gov/genetics-glossary/Transcription

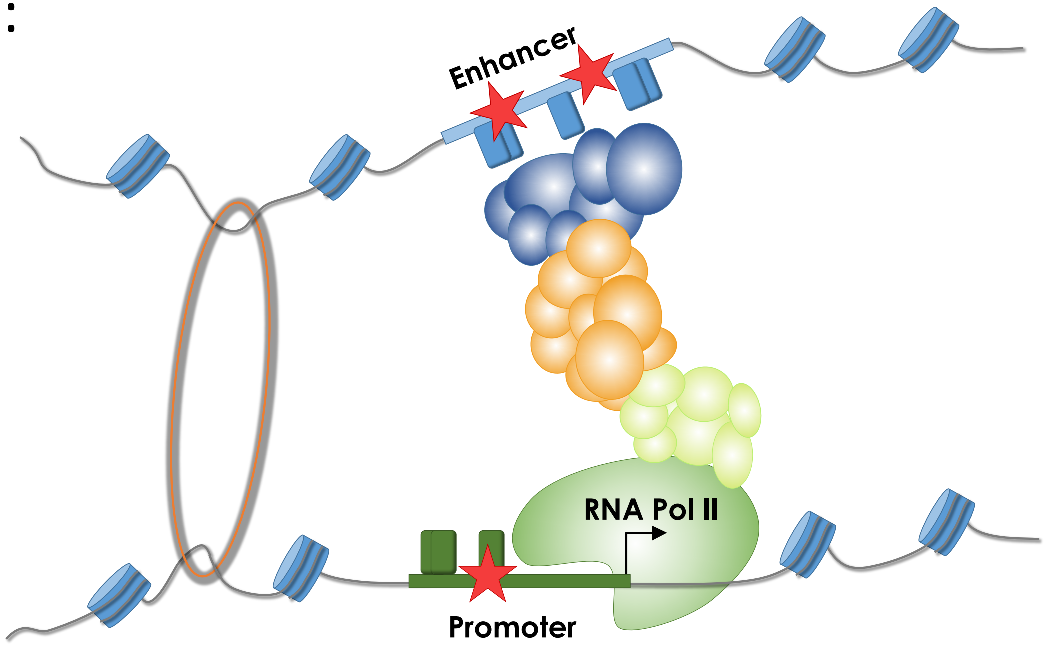




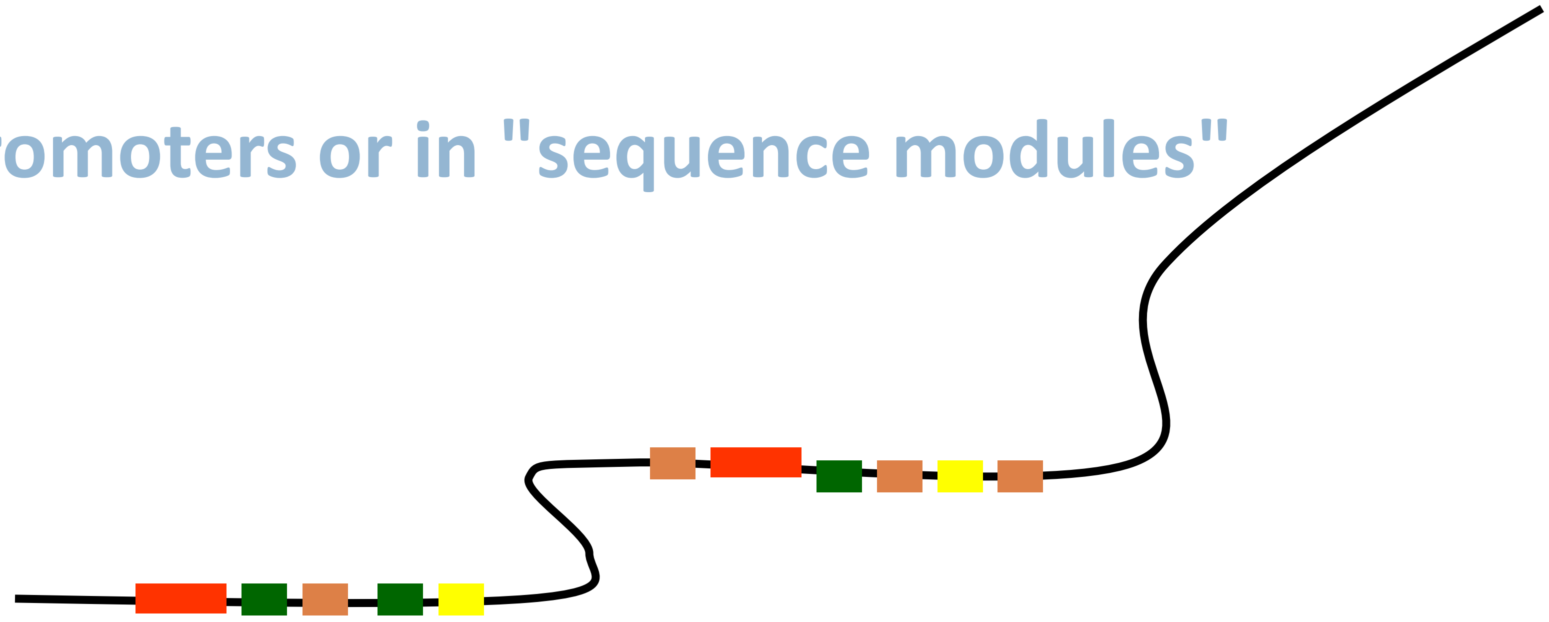
https://www2.lbl.gov/Science-Articles/Archive/sabl/2005/March/assets/proteins_as_galaxies.jpg

Two classes? Coding vs. Non-coding variants?

- **Non-coding sequences**: ~98% of the genome remaining after subtracting coding sequence
- **Functional non-coding elements**: like 3' and 5' UTRs, short and long RNA species, …
- **Regulatory sequences** (~5-15% of genome): enhancer, silencer, promoter, insulator
  - Open chromatin regions?
  - Chromatin associated?



TFBSs are clustered in promoters or in "sequence modules"



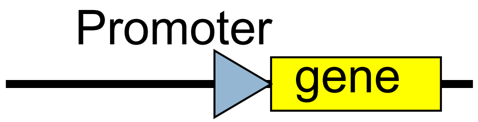
The distribution of binding sites in the genome is non-uniform

In small genomes, most sites are in promoters, and there is a bias toward a nucleosome free region near transcriptional start sites (TSSs)

In larger genomes (fly) CRMs (cis-regulatory-modules) exist which are frequently away from the TSS (enhancers)

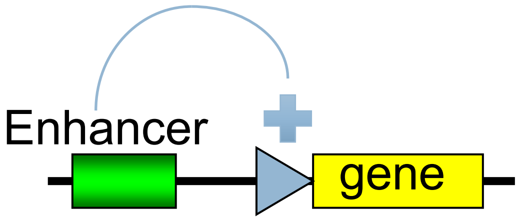
A single binding site, without the context of other TF, is unlikely to represent a functional loci

Operational definitions of cis-regulatory modules



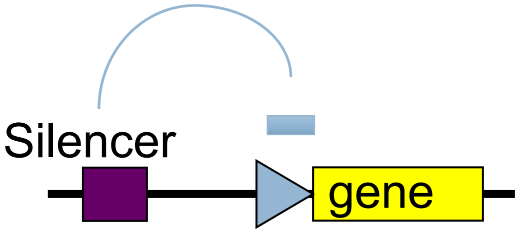
A promoter is the DNA sequence required for correct initiation of transcription

5′ end of the gene (may overlap exons/coding sequence)



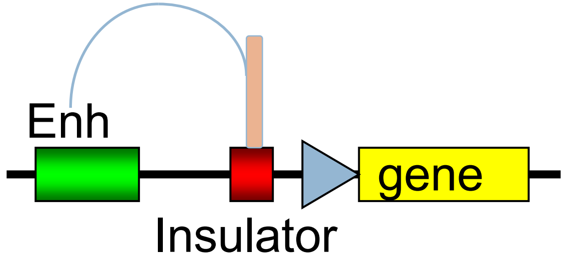
An enhancer is a DNA sequence that causes an increase in gene expression

Act independently of position and orientation with respect to the gene



A silencer is a DNA sequence that causes a decrease in gene expression

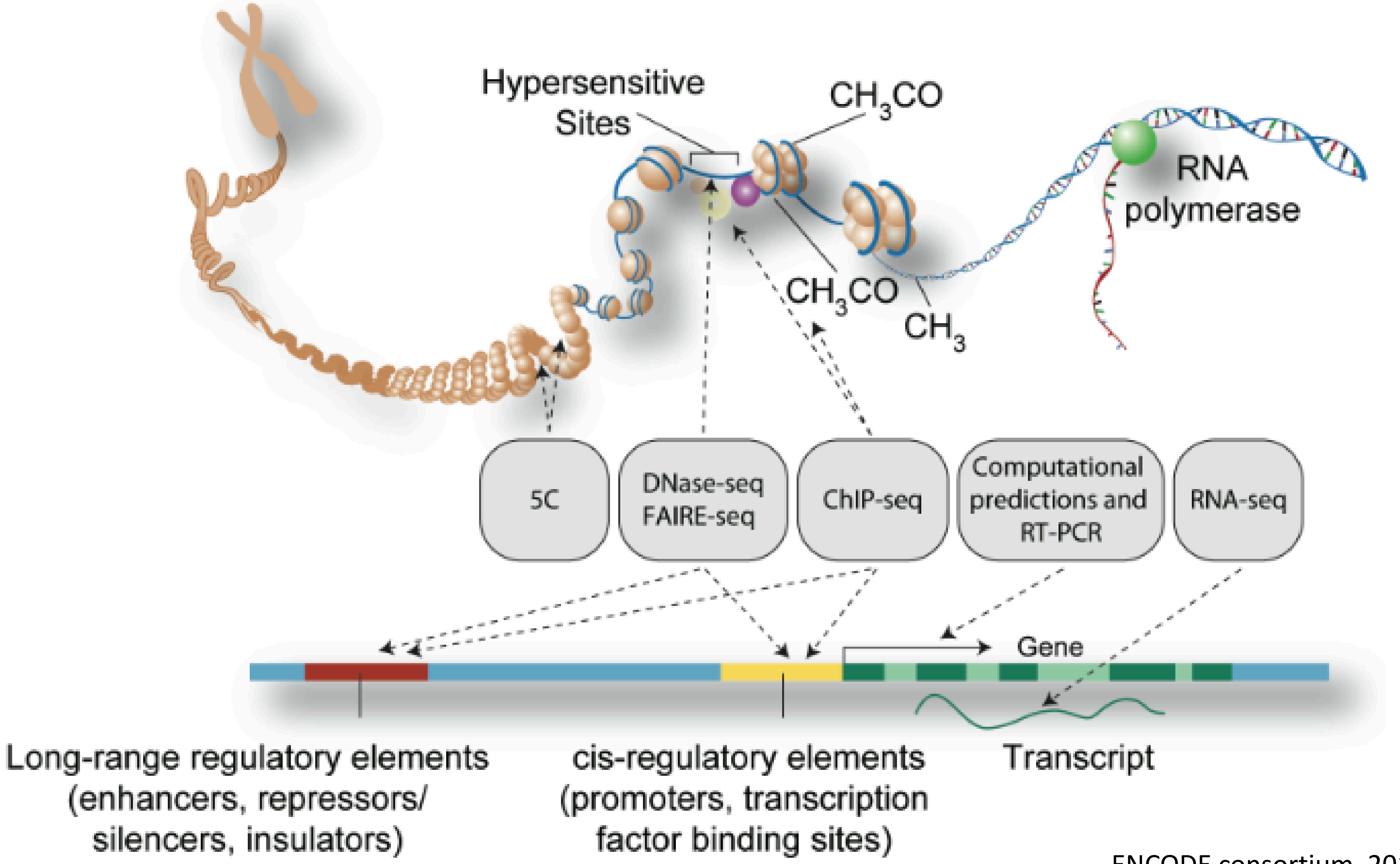
Similar to enhancer but has an opposite effect on gene expression (can be physically the same sequence and depend on cell-type)



An insulator is a DNA sequence that blocks activation of a promoter by an enhancer

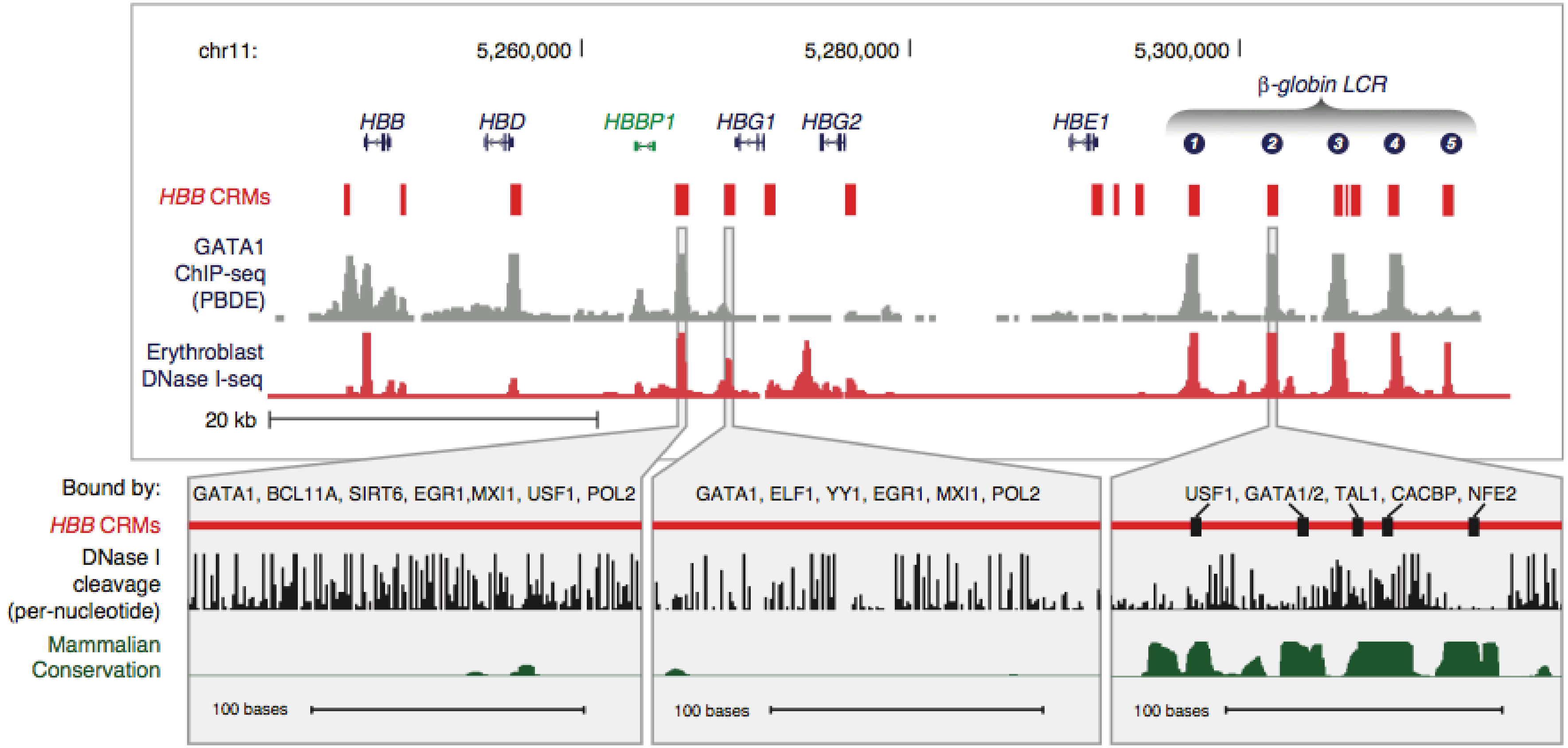
Formed by topologically associated domains (TADs) by CTCF and Cohesin binding

Mapping Functional Elements



ENCODE consortium, 2011, PLoS Biology 9: e1001046

Diversity of evidence supporting regulatory regions



Evidence for function:	*HBG1*-D BCL11a	*HBG1* 3' enhancer	β-globin LCR HS2
Genetics	–	+	+
Biochemical	+	+	+
Evolution	–	–	+

Jeff Vierstra, John Stamatoyannopoulos, R Hardison in Kellis et al. 2014 PNAS

What is biological function?

"A genome segment may be biochemically active yet have no biologically meaningful function. An analogous situation arises when my shoe binds a piece of chewing gum during hot days in Houston. The binding of the chewing gum is an observable activity, but no reasonable person would claim that this is the function of shoes."

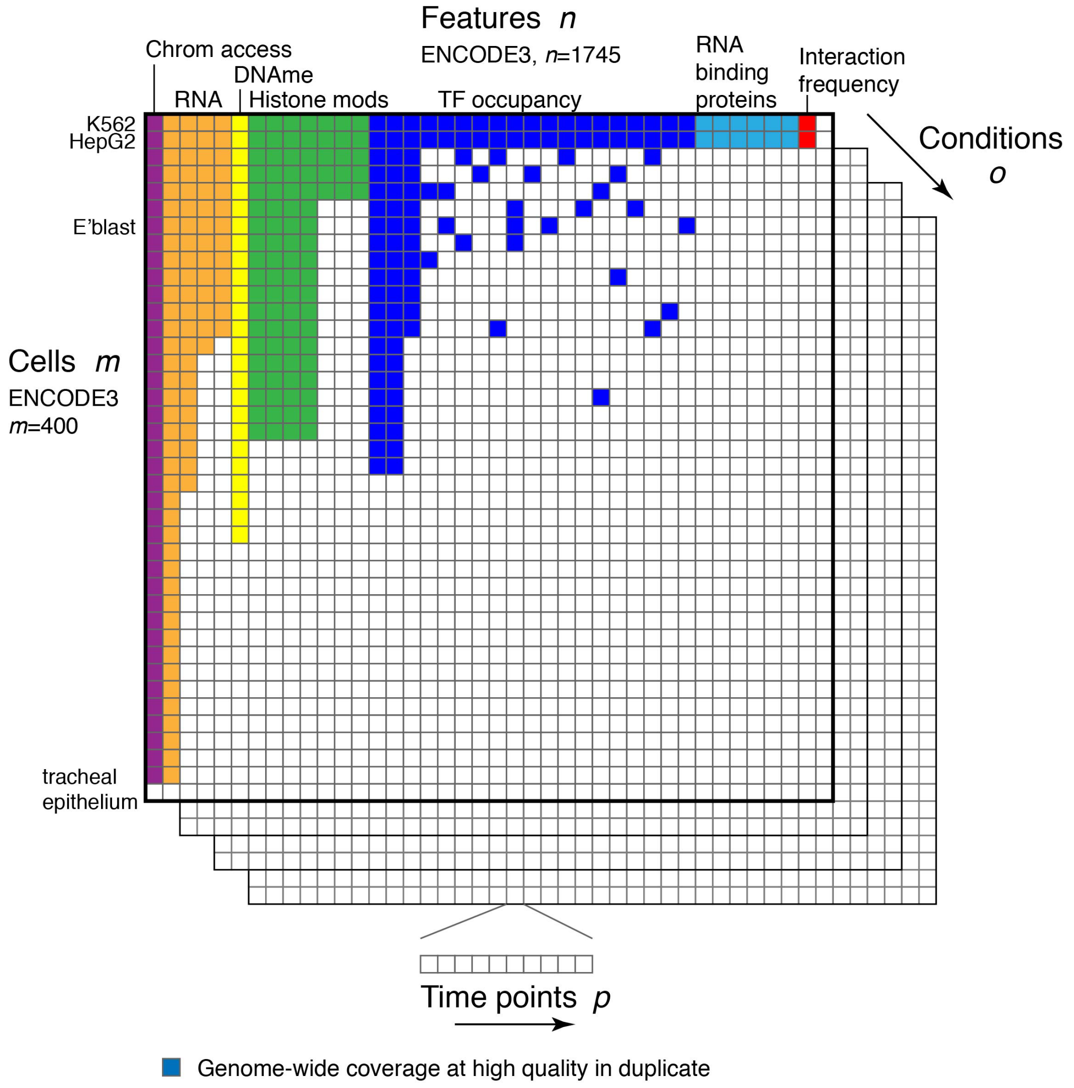
Dan Graur, 2016

The current state of mapping function-associated features

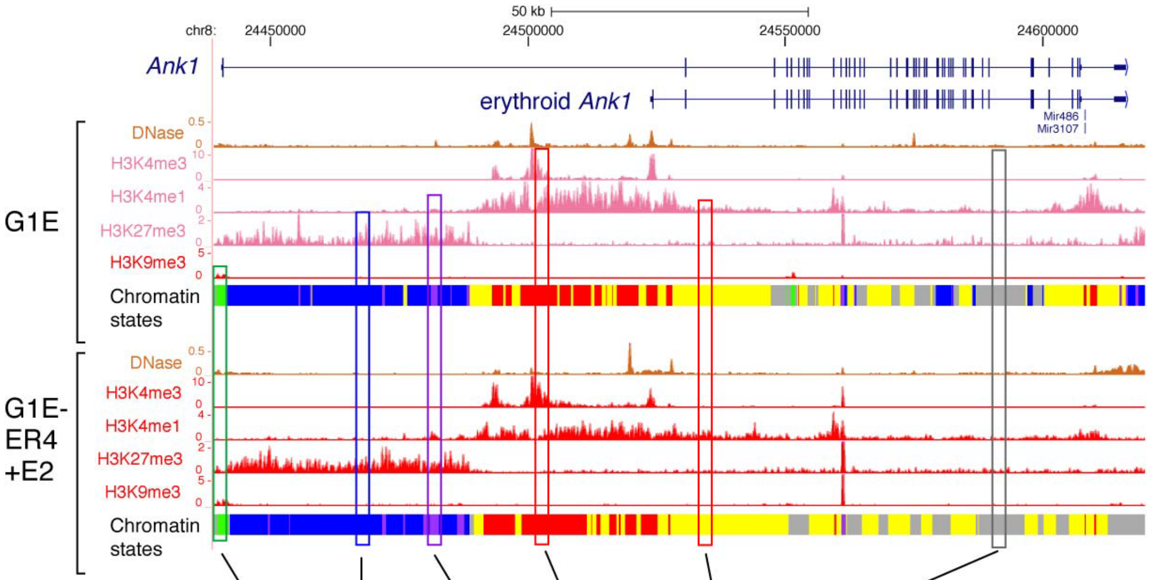
Thousands of experiments, but sparse across diversity of cell-types, conditions or developmental trajectories

Computational challenges:

- **Data integration (batch/ technical confounders)**
- **Data imputation**
- **Catalogs of elements**

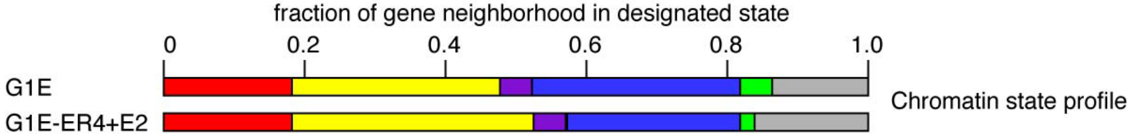


"Paint" genomic regions by dominant histone modifications: multivariate HMM (chromHMM)



State	5	4	3	1	2	6
H3K4me1	0.1	0.5	79.9	70.3	75.0	0.1
H3K4me3	0.9	0.0	9.5	97.8	0.2	0.0
H3K27me3	1.5	42.6	78.9	0.9	0.4	0.2
H3K9me3	72.7	0.2	6.3	0.7	0.5	0.2
Input	0.8	1.1	3.9	0.7	1.1	0.2

Emission frequencies of each state in HMM

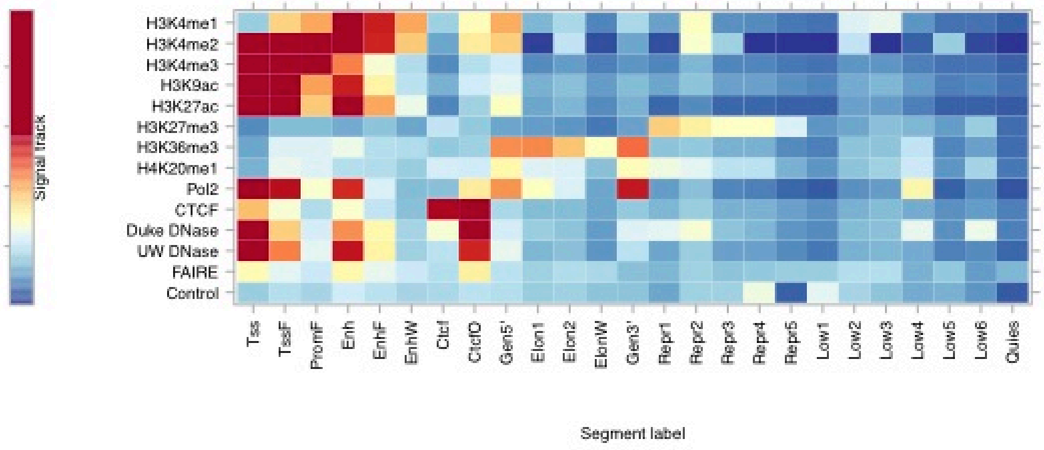


Ernst and Kellis (2010) Nature Biotechnology 28: 817… Wu et al. 2011 **Genome Res** *21*: 1659-1671

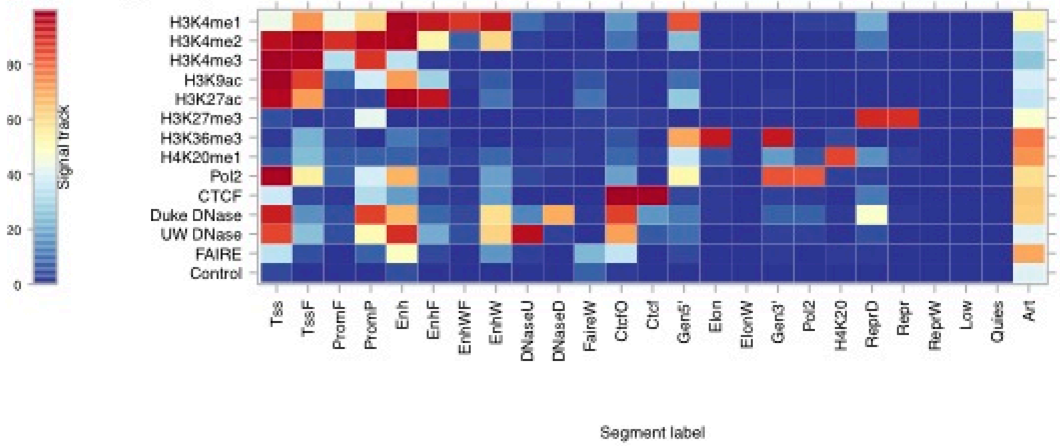
Integrate features using multivariate segmentations (Segway)

Regul	Regulatory
Tss	Transcription start site
Prom	Promoter
Gen5′	5′ ends of genes
Elon	Elongation
Gen3′	3′ ends of genes
Enh	Enhancer
EnhPr	Enhancer/Promoter
Ctcf	CTCF binding
Dnase	DNaseI sensitivity
Faire	FAIRE activity
Pol2	Pol2 binding
H4K20	H4K20 monomethylation
Repr	Repressed
Low	Low signal
Art	Artifact
Quies	Quiescent
W	Weak
F	Flanking
P	Poised
O	Open
D	Duke
U	U Washington

(A) Label mnemonics



(B) Segway



(C) ChromHMM

ENCODE Project Consortium (2012) Nature

M. Hoffman, J. Ernst et al. 2013. Integrative segmentations of function-associated marks. NAR

Catalogs of open chromatin regions

https://screen.wenglab.org/

EH38E2941902 chr11:5,267,726-5,267,987 D

In Specific Biosamples | Nearby Genomic Features | TF and His-mod Intersection | TF Motifs and Sequence Features | Associated Gene Expression | Associated RAMPAGE Signal | Linked cCREs in other Assemblies | Functional Data | EN-TEx States | Signal Profile | Linked Genes

Cell type agnostic classification

Search:

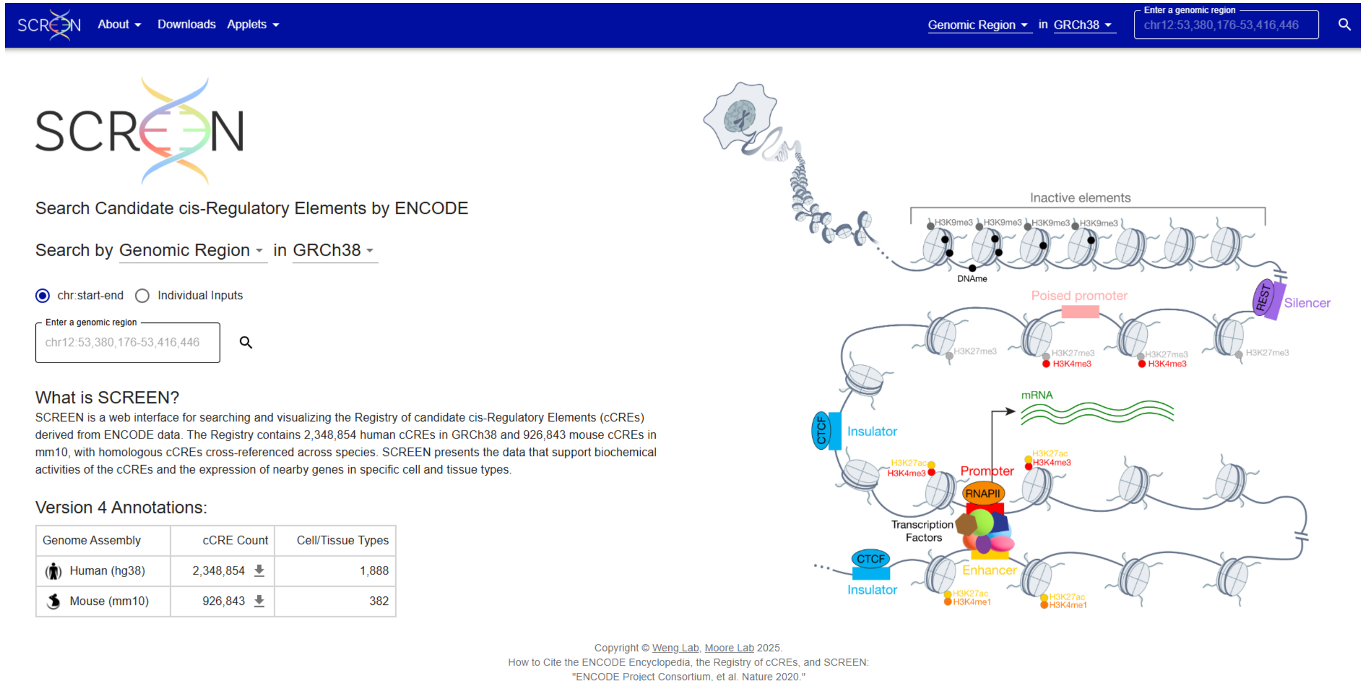
Cell Type	DNase max-Z	H3K4me3 max-Z	H3K27ac max-Z	CTCF max-Z	Group
cell type agnostic	1.68	2.19	2.02	1.45	distal enhancer-like signature

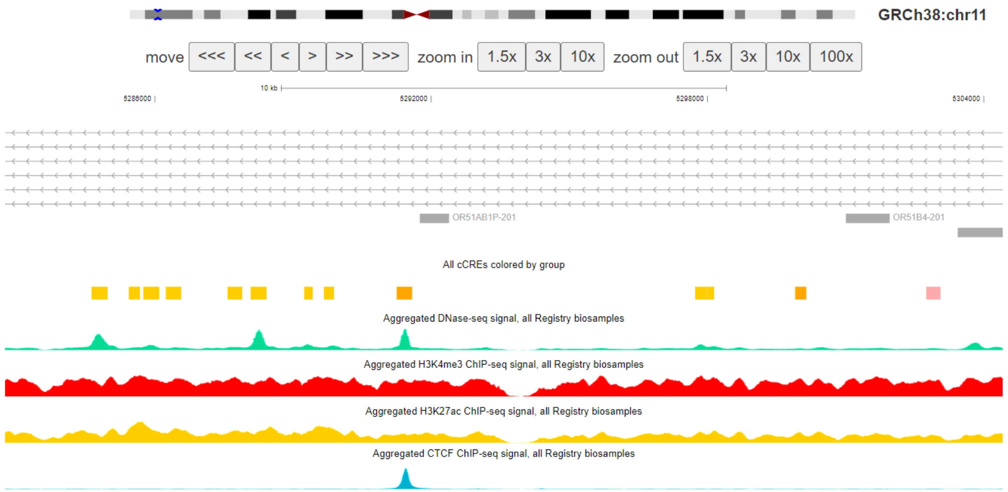
Total: 1

Classification in Type A biosamples (all four marks available)

Search:

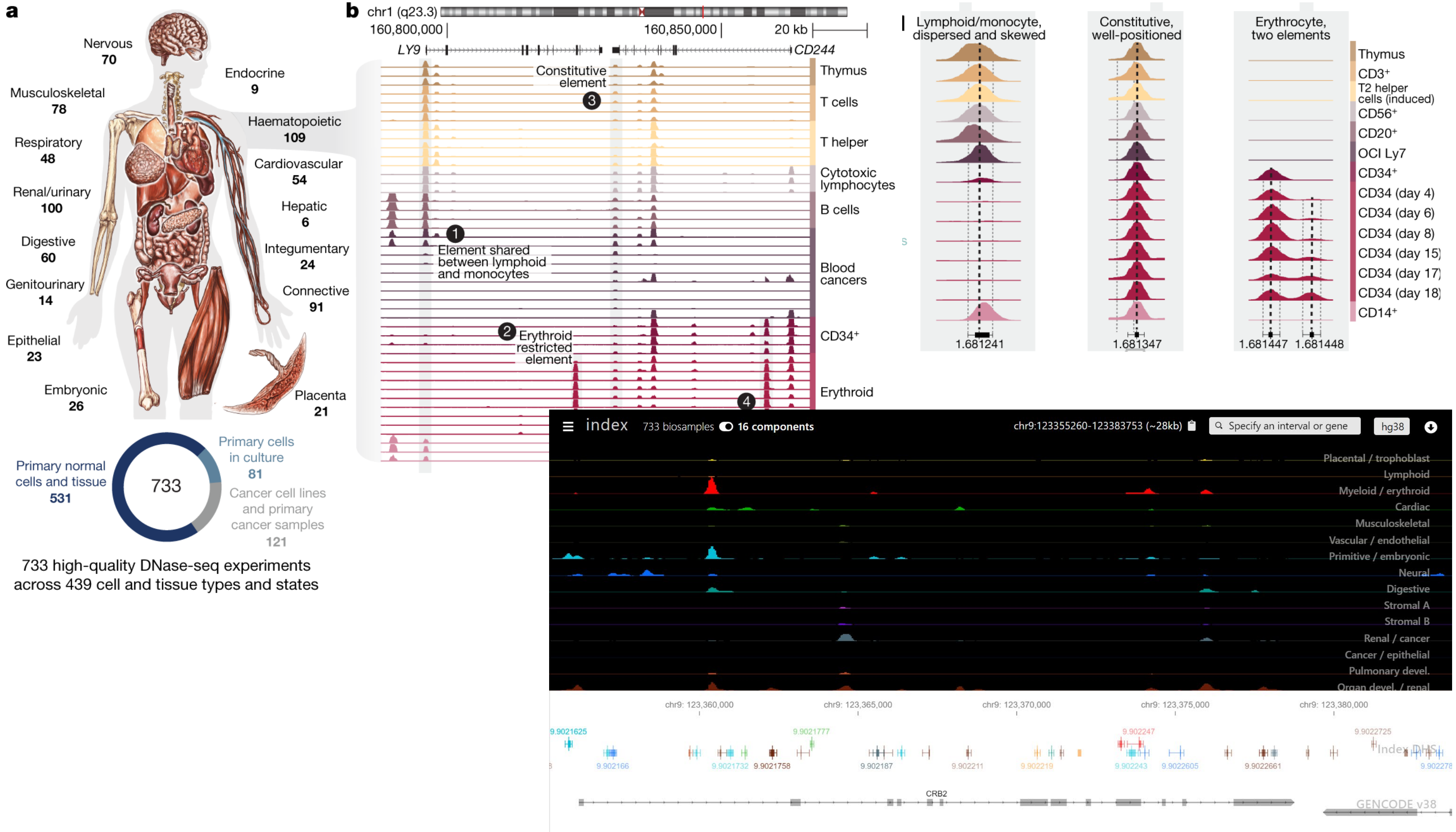
cell type	DNase Z-score	H3K4me3 Z-score	H3K27ac Z-score	CTCF Z-score	Group
K562	1.68	2.19	2.02	-0.01	distal enhancer-like signature
keratinocyte female donor ENCDO268AAA	0.54	0.16	-10.00	-10.00	low DNase





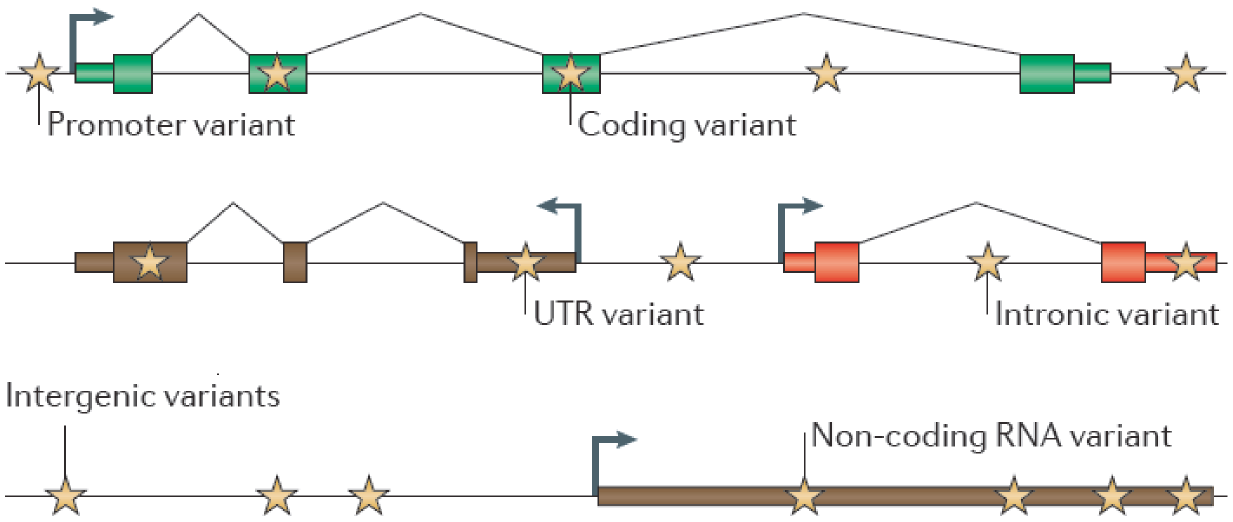
Catalogs of open chromatin regions (2)

https://index.altius.org/



Multiplex Assays of Variant Effects (MAVE)

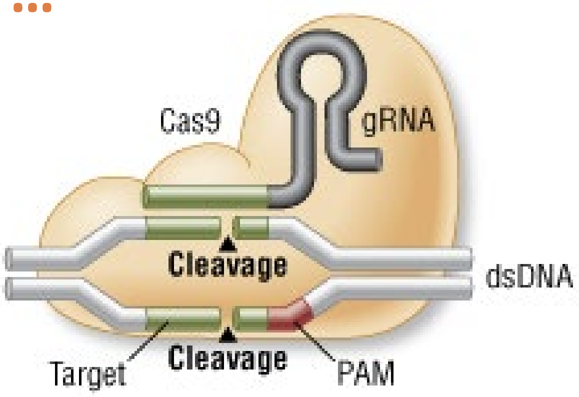
- **A toolbox for interrogating the effects of sequence alterations**
- **Diverse molecular assays**
  - Lenti-virus, piggyBac transposon, …
  - Directed or random mutagenesis
  - CRISPR/Cas and dCas fusions
- **Diverse readouts**
  - Survival / proliferation
  - Protein abundance
  - Reporter signal (e.g. light), RNA expression, DNA and chromatin state



Modified from Cooper and Shendure, *Nature Review Genetics* 2011

Quantitative read-outs from regulatory sequences

- **CRISPR/Cas9: mutation, deletion, activator/repressor screens, …**
- **Massively Parallel Reporter Assays (MPRAs) or similar:**
  - Test activity of regions (element catalogs / rules)
  - Measuring effects of genomically scattered mutations
  - Dense read outs for mutations in select regions

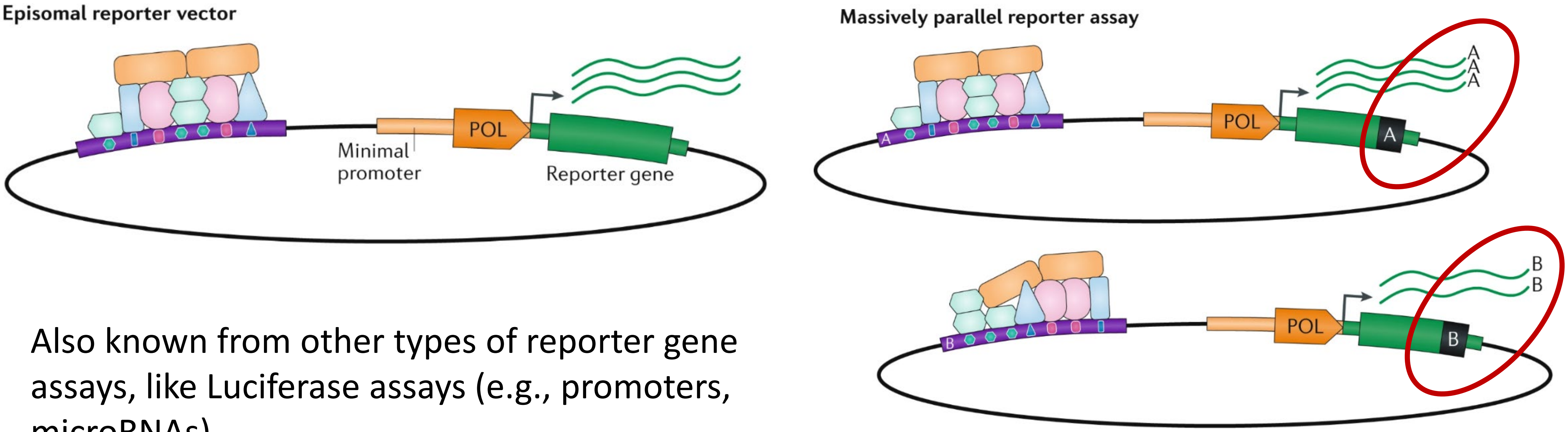


www.neb.com

	Construct	Tested in	Detection	Advantages	Disadvantages
MPRA/ MPFD/ CRE-seq	Enhancer – P – Reporter – BC – pA	Cell lines, Mouse liver, Mouse retina	Barcode RNA-seq	High BC multiplicity Quantitative	Episomal
STARR-seq	P – Reporter – Enhancer – pA	Cell lines	Enhancer RNA-seq	Quantitative	Low multiplicity Episomal
TRIP	5'TR – P – Reporter – BC – pA – 3'TR	Mouse ESCs	Barcode RNA-seq	Quantitative Genomic context	Low resolution

F. Inoue & N. Ahituv, Genomics 106(3), September 2015, Pages 159-164

Massively parallel reporter assays (MPRAs)



Also known from other types of reporter gene assays, like Luciferase assays (e.g., promoters, microRNAs)

Gasperini et al, *Nat Reviews Genetics* 2020

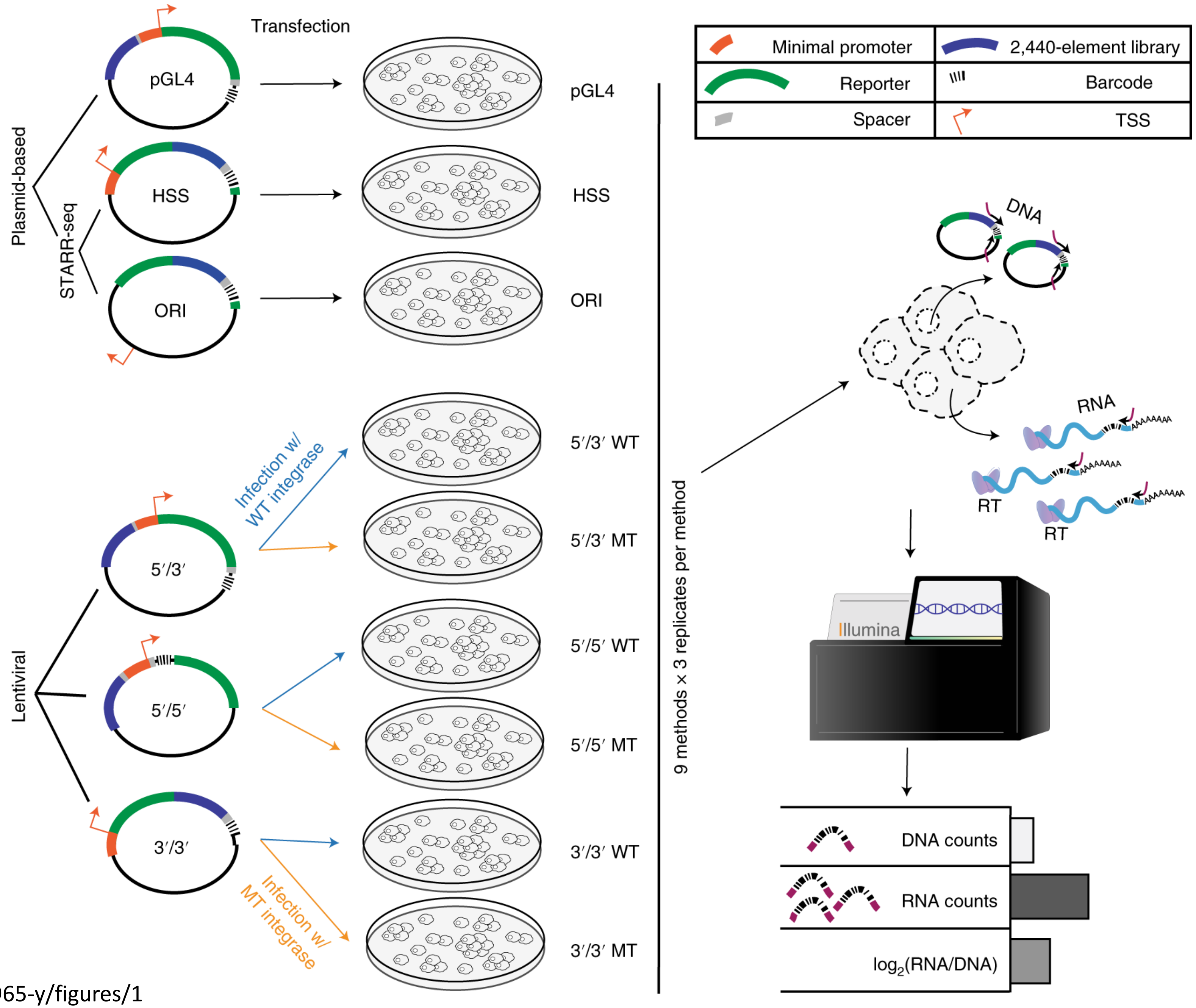
MPRA "every way"

nature methods ANALYSIS
https://doi.org/10.1038/s41592-020-0965-y
Check for updates

A systematic evaluation of the design and context dependencies of massively parallel reporter assays

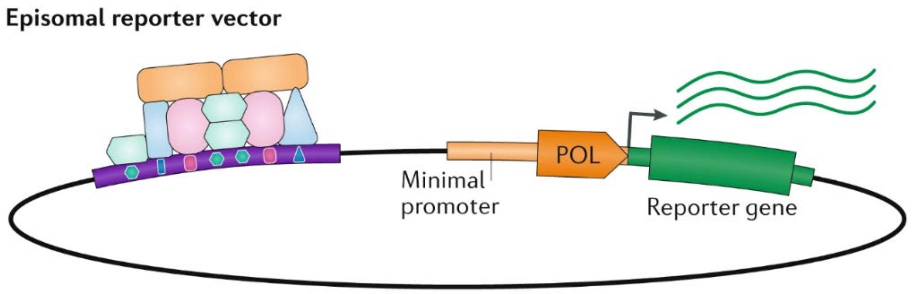
Jason C. Klein[1,9,11], Vikram Agarwal[1,2,11], Fumitaka Inoue[3,4,10,11], Aidan Keith[1,11], Beth Martin[1], Martin Kircher[1,5,6], Nadav Ahituv[3,4] ✉ and Jay Shendure[1,7,8] ✉

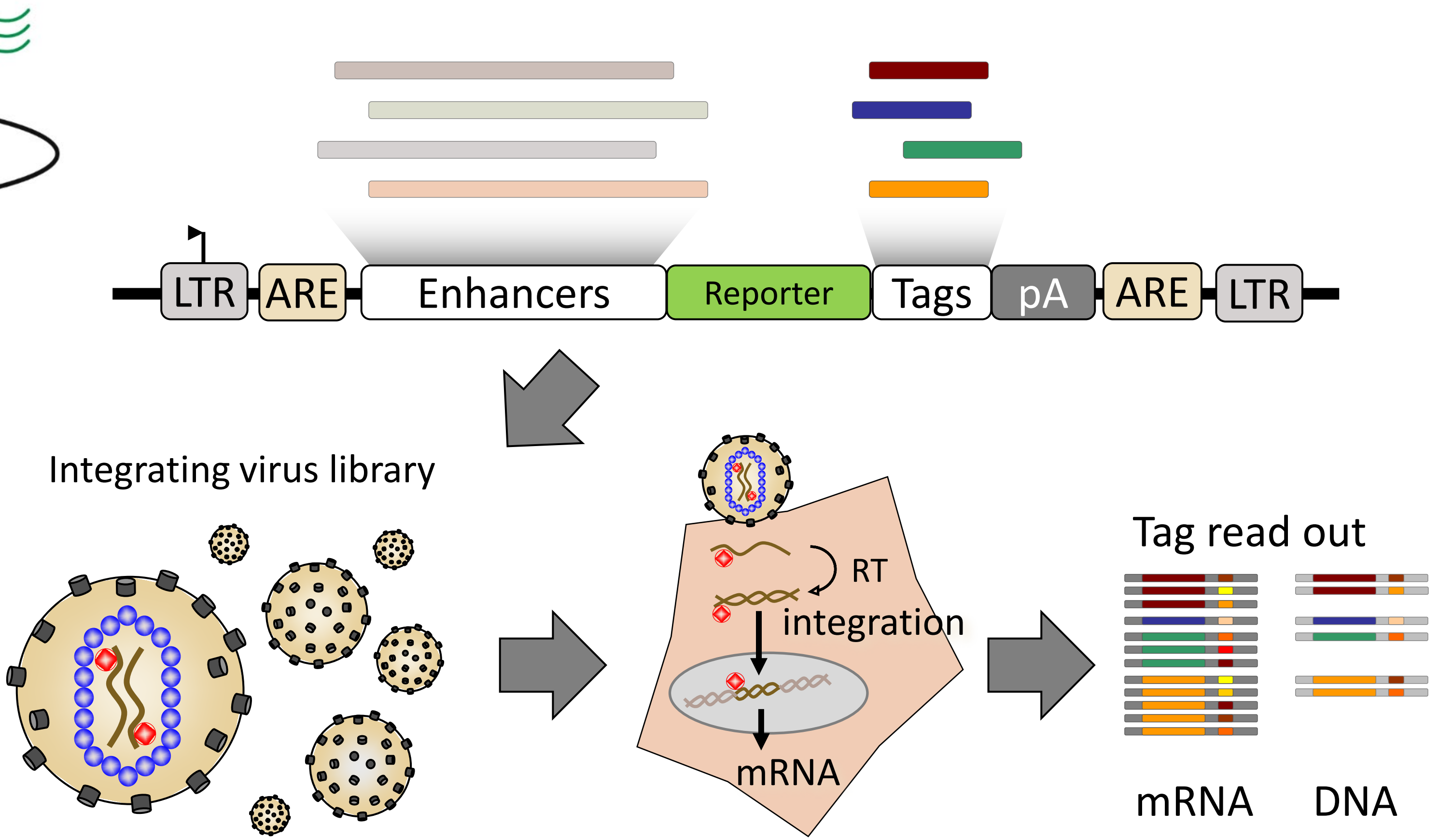
Massively parallel reporter assays (MPRAs) functionally screen thousands of sequences for regulatory activity in parallel. To date, there are limited studies that systematically compare differences in MPRA design. Here, we screen a library of 2,440 candidate liver enhancers and controls for regulatory activity in HepG2 cells using nine different MPRA designs. We identify subtle but significant differences that correlate with epigenetic and sequence-level features, as well as differences in dynamic range and reproducibility. We also validate that enhancer activity is largely independent of orientation, at least for our library and designs. Finally, we assemble and test the same enhancers as 192-mers, 354-mers and 678-mers and observe sizable differences. This work provides a framework for the experimental design of high-throughput reporter assays, suggesting that the extended sequence context of tested elements and to a lesser degree the precise assay, influence MPRA results.



https://www.nature.com/articles/s41592-020-0965-y/figures/1

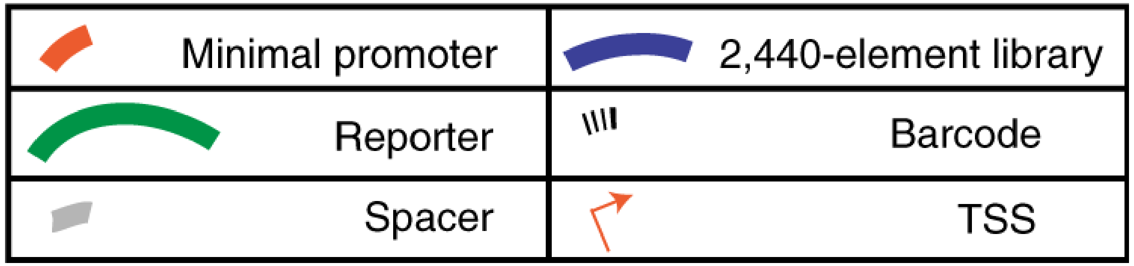
Episomal vs. lentiMPRA genomic integration

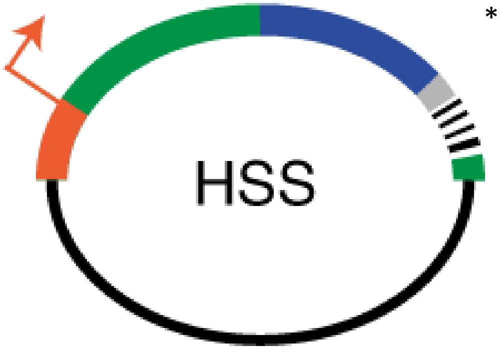




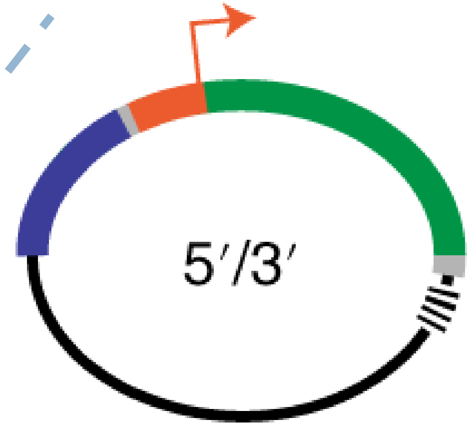
doi: 10.1101/gr.212092.116

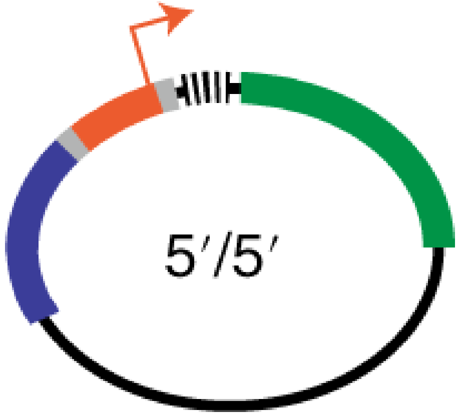
STARR-seq vs barcoded constructs



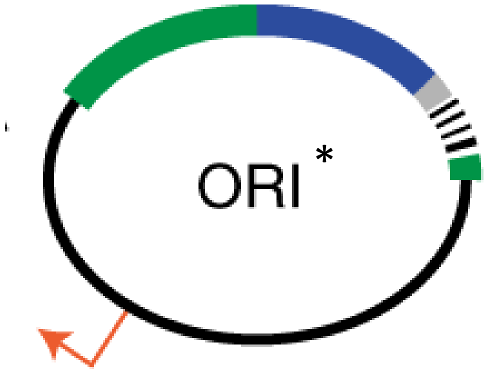


* https://pubmed.ncbi.nlm.nih.gov/29256496/

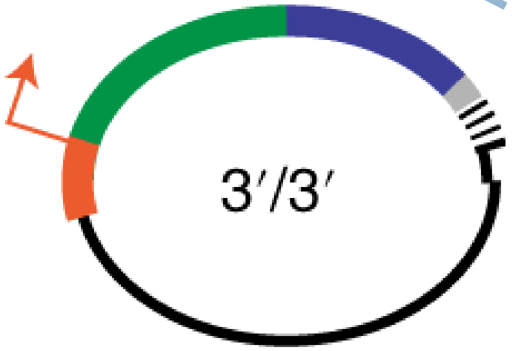


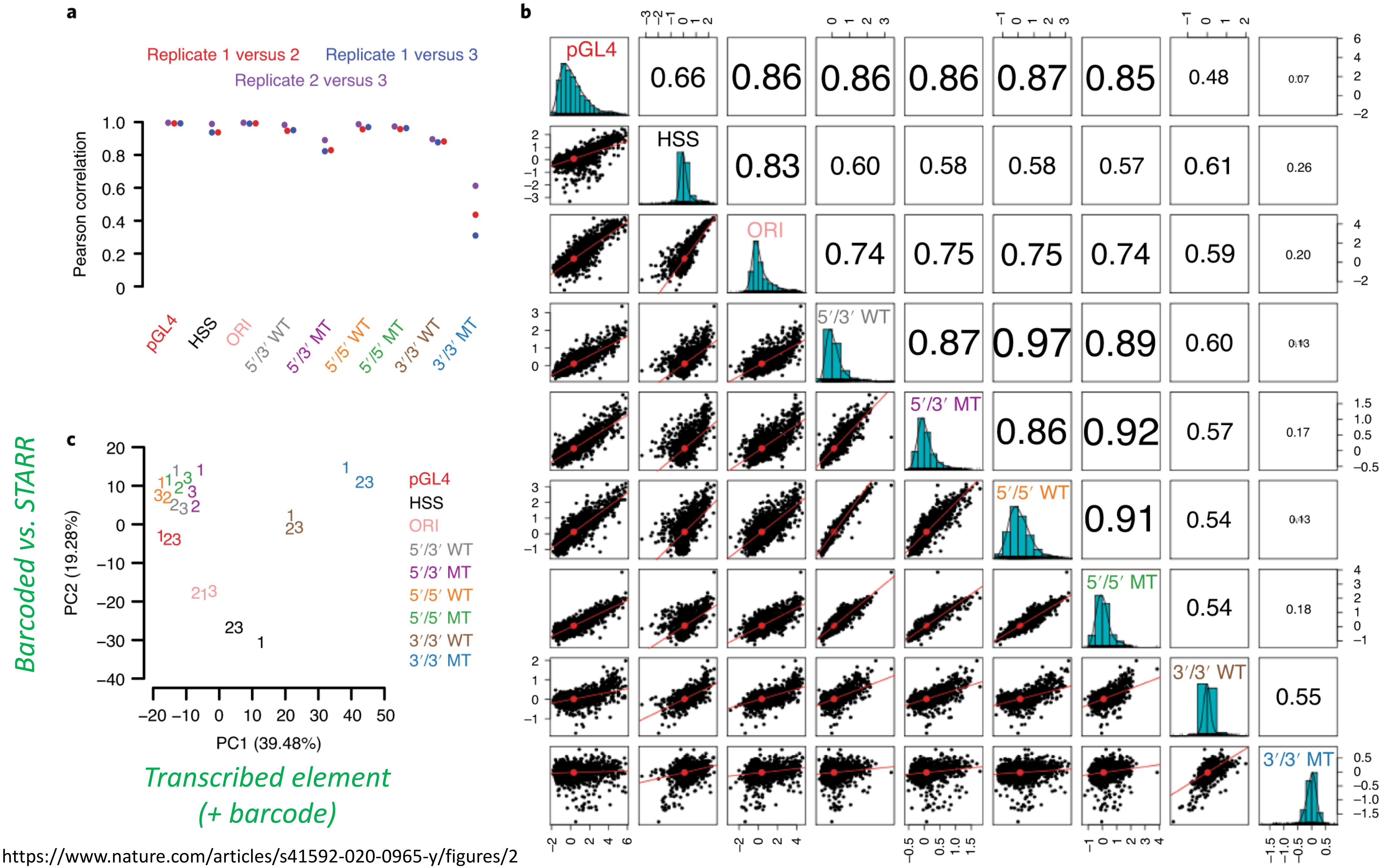


STARR-seq – tested sequence is being transcribed and identified through RNA sequencing



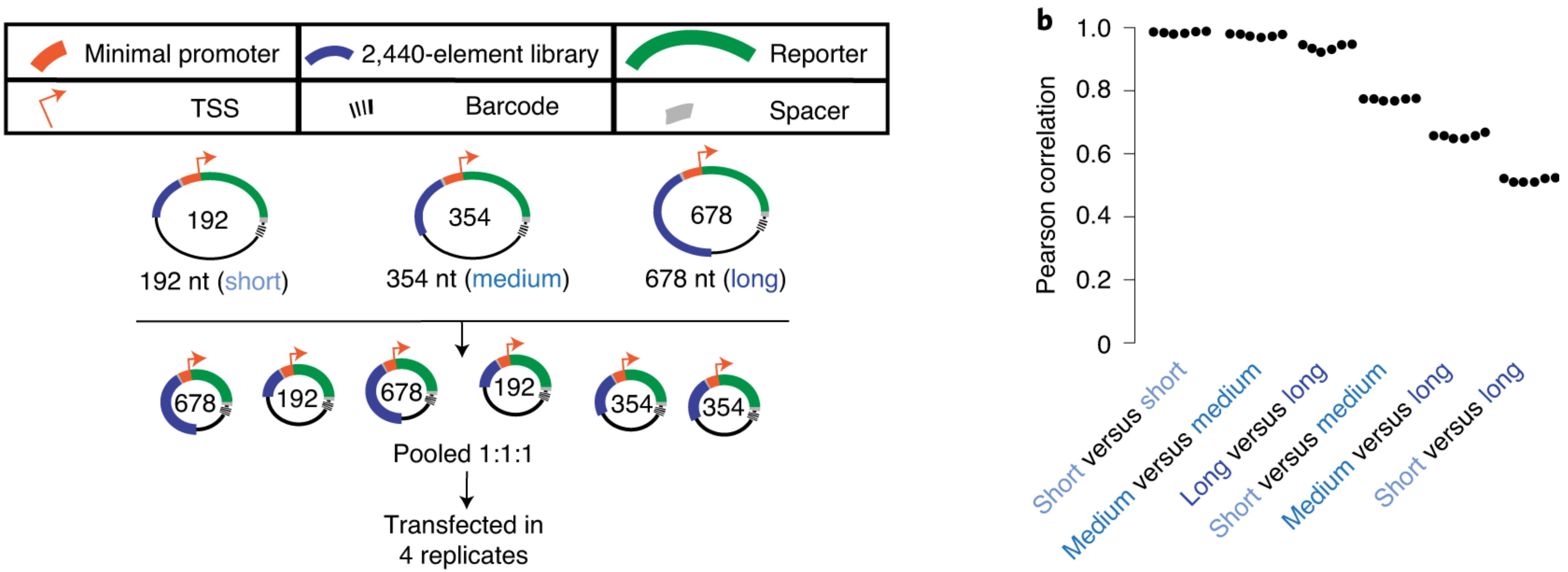
CRE-seq/barcoded constructs – tested sequence is represented by a (short) transcribed barcode





https://www.nature.com/articles/s41592-020-0965-y/figures/2

Length of the tested sequence has an effect



Testing longer sequences comes with more replicate variation, but longer sequences are adding biologically relevant signal.

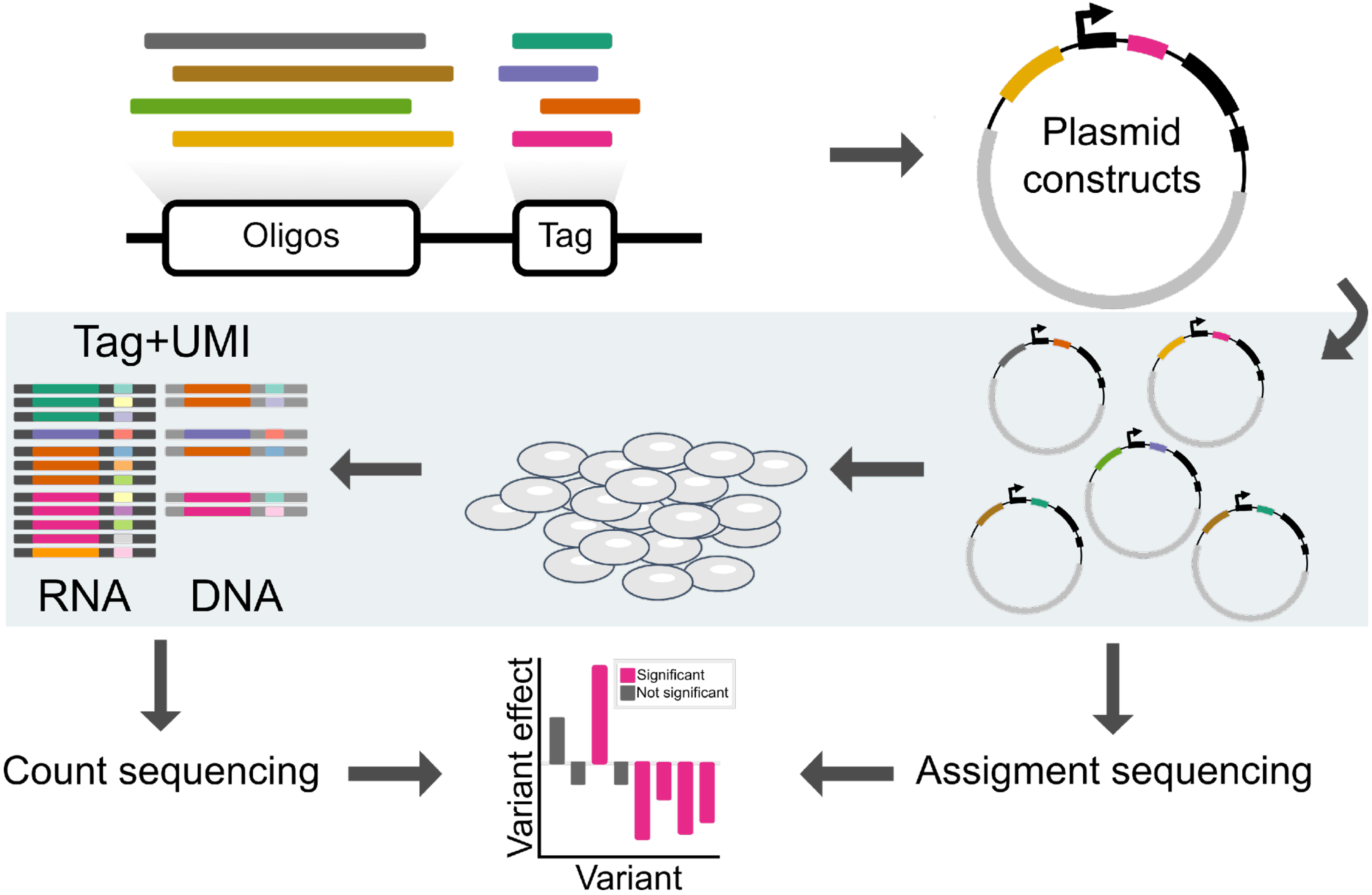
https://www.nature.com/articles/s41592-020-0965-y/figures/5

Overview of challenges with MPRAs

- **Artificial nature (e.g., reporter, cell lines, …)**
- **Experimental complexity (e.g., RNA/DNA extraction, working with complex DNA libraries, concordance between experimentalists/labs…)**
- **Variety of experimental designs (e.g., vector, targets, barcodes, UMIs, …)**
- **Typically, length of tested sequences is constrained, e.g. by synthesis**
- **Elements are easier than variants**
  - Variants require successful measurements for at least two alleles
  - Detection of variant effect will depend on activity of its element
  - Element effects can be modeled much better – reaching MPRA replicate correlations
- **The more sequences (the higher the complexity) the more difficult the experiment**

Part II – What’s the output and how to process the data?

Massive Parallel Reporter Assay (MPRA)



FastQ files

```
@SOLEXA-GA03_0006_PEi_DH_FD_KG_VW_SP:5:1:1033:5267
AGACAGACACAGAGNAAGACCCAGTCCGCCACACAGGCAAACTCACGCAGTACGCGCCGCAGCGGGAGAGACACAAACCGTAGTGCCTCTGCCTGTGTCTCTCTGCCTCTGTCTCTGCCTCTGTCTCTCTCTGCCTCTGTCTCTGTCTCTGTCTCTGCC
+
4--'-(/.23/044!51/+//.400/-/1-62/.6021834///62126313/2230+2/31/3113224/+2//.938643153256956368576789;979844:;797786984778:8:68595789943:5::81:0885/61334/5+10--
@SOLEXA-GA03_0006_PEi_DH_FD_KG_VW_SP:5:1:1033:18417
CAGCATGCTGGTCGNCCCACCTCAGGGCCGAAGGCACCCATGCACCTGGCAGGTTGGCGTCCCCAGTGCTGAGGAAATGGTATCCATTATGATCTGGATCTATGGAGGCGCCTTCCTCATGGGGTCCGGCCACGGGGCCAACTTCCTCAACAACTACCT
+
45628576753499!37447349;726458:82849745;:998669:46:965663365333336723.2133221536543752688898;=9:;8<=9<==96;:87988:97697;:6676645836486755514960863340461462540.0
```

⋮

Read number
Quality pass
Control bit
Index1+Index2

Instrument coordinates

RunID without date

lane tile X Y

Sequence identifier

```
@SOLEXA-GA03_0006_PEi_DH_FD_KG_VW_SP:5:1:1033:5267 2:N:0:XXXXX+YYYYY
AGACAGACACAGAGNAAGACCCAGTCCGCCACACAGGCAAACTCACGCAGTACGCGCCGCAGCGGGAGAGACACA
+
4--'-(/.23/044!51/+//.400/-/1-62/.6021834///62126313/2230+2/31/3113224/+2//
```

Sequence

Quality score line

Phred* quality score

Given e, the estimated probability of the base call being wrong:

$$Q = -10 \log_{10} e$$

Higher Q scores indicate a smaller probability of error

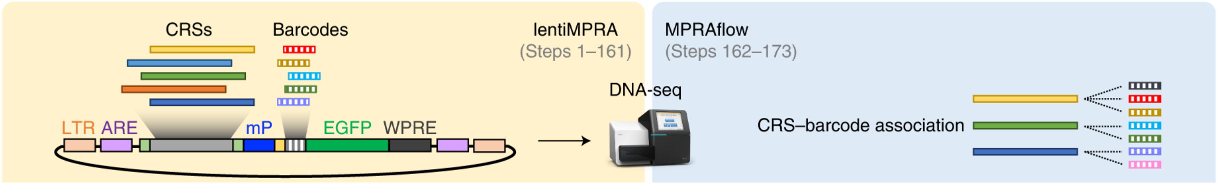
Example: A quality score of 20 represents an error rate of 1 in 100, i.e., a call accuracy of 99%; a quality score of 30 corresponds to an error rate of 1/1000 (accuracy of 99.9%)

Enconding usually as ASCII char 33 to 74 (depending on technology) for a Phred quality score from 0 to 41.

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	Space	64	40	@	96	60	`
33	21	!	65	41	A	97	61	a
34	22	"	66	42	B	98	62	b
35	23	#	67	43	C	99	63	c
36	24	$	68	44	D	100	64	d
37	25	%	69	45	E	101	65	e
38	26	&	70	46	F	102	66	f
39	27	'	71	47	G	103	67	g
40	28	(	72	48	H	104	68	h
41	29	)	73	49	I	105	69	i
42	2A	*	74	4A	J	106	6A	j
43	2B	+	75	4B	K	107	6B	k
44	2C	,	76	4C	L	108	6C	l
45	2D	-	77	4D	M	109	6D	m
46	2E	.	78	4E	N	110	6E	n
47	2F	/	79	4F	O	111	6F	o
48	30	0	80	50	P	112	70	p
49	31	1	81	51	Q	113	71	q
50	32	2	82	52	R	114	72	r
51	33	3	83	53	S	115	73	s
52	34	4	84	54	T	116	74	t
53	35	5	85	55	U	117	75	u
54	36	6	86	56	V	118	76	v
55	37	7	87	57	W	119	77	w
56	38	8	88	58	X	120	78	x
57	39	9	89	59	Y	121	79	y
58	3A	:	90	5A	Z	122	7A	z
59	3B	;	91	5B	[	123	7B	{
60	3C	<	92	5C	\	124	7C	|
61	3D	=	93	5D	]	125	7D	}
62	3E	>	94	5E	^	126	7E	~
63	3F	?	95	5F	_	127	7F	□

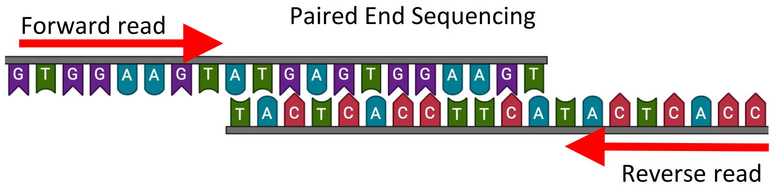
* Named after a software that introduced quality score calculation for Sanger sequencing

Output data from association or assignment sequencing



Main sequencing reads

- Single or Pair of FASTQ file(s) of the target

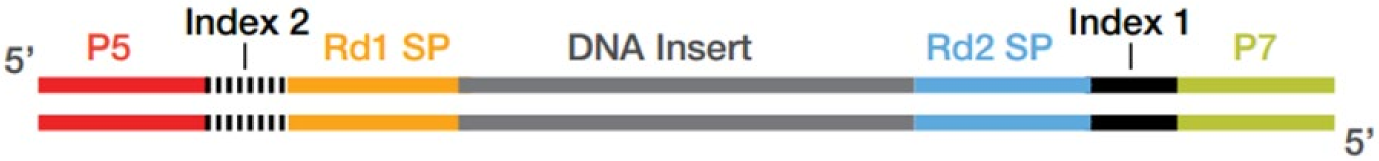


Target barcode

- Typically, separate FASTQ file (for i5 and/or i7 index)

Sample index

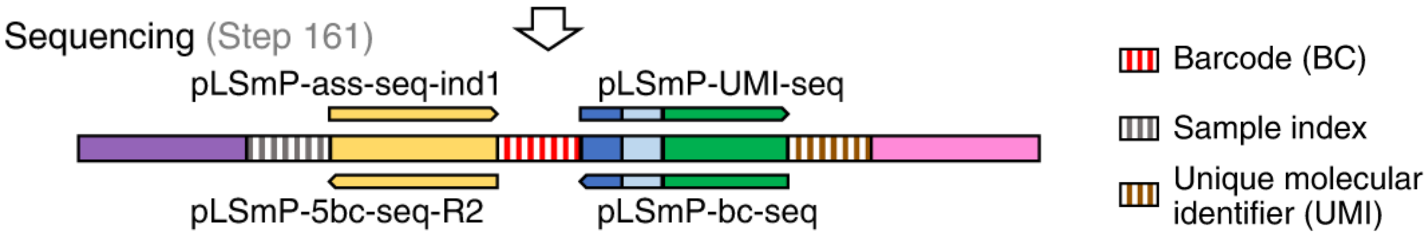
- Typically, separate FASTQ file (for i5 and/or i7 index), most likely already demultiplexed by sequencing core



https://support.illumina.com/bulletins/2020/06/illumina-adapter-portfolio.html

Index, barcode and UMI sequences

TAACAGCCAA
CTAAGAGTCC
TTACTGCCTT
CGCTGAATTC
TGACGTCCTT
TGTGTGTAAC
AGATTGAGAG
GATAATGATG
TTGAAGAAGC
CTGATACTTC



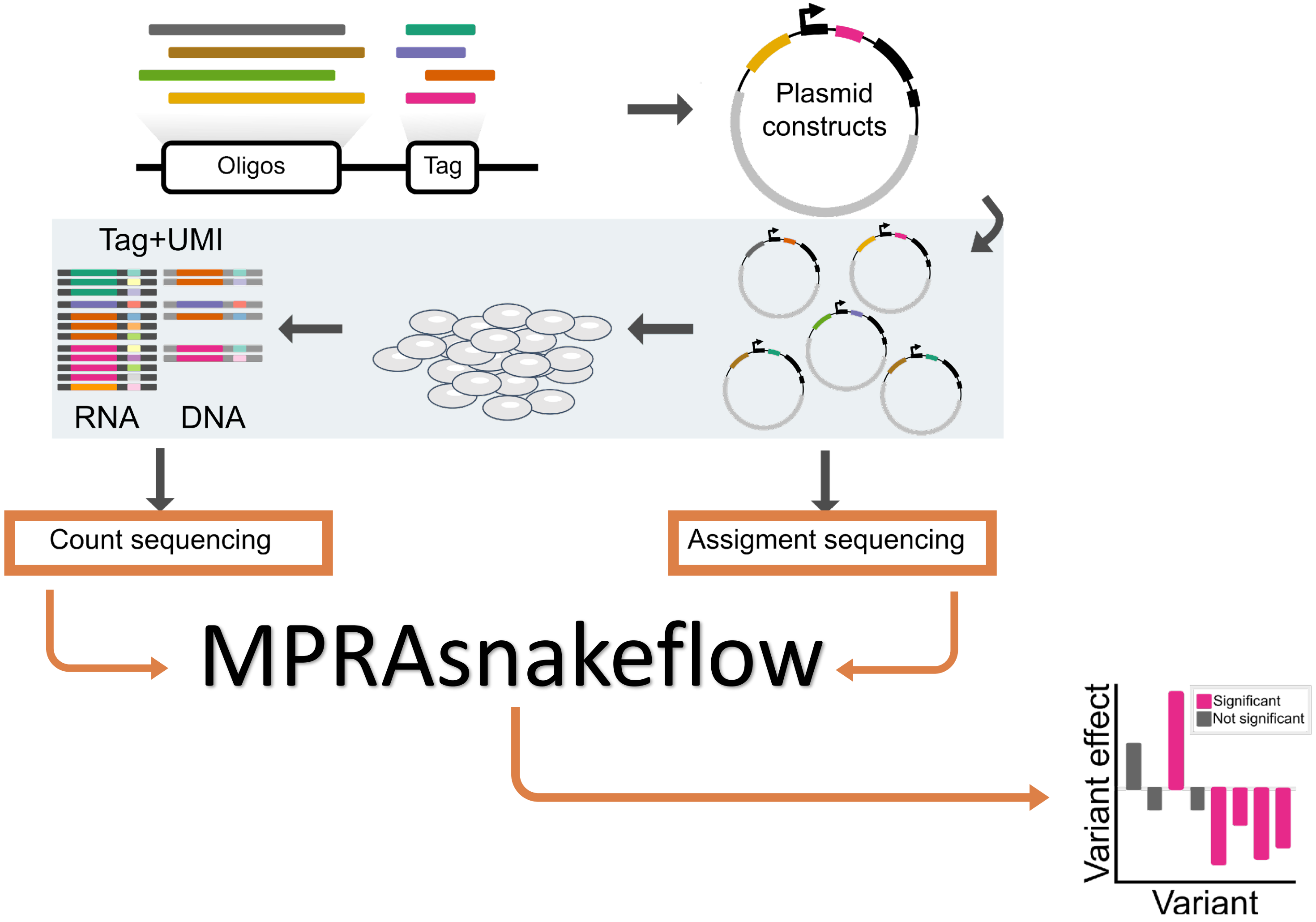
… are short sequences used to uniquely tag a sample, partition/cell/targets or single molecules in an experiment

Typically,

- Indexes tag samples (e.g., replicates)
- Barcodes tag targets (e.g., counts)
- UMIs (unique molecular identifiers) tag molecules before amplification (for count estimation and error correction)

https://support.illumina.com/bulletins/2020/06/illumina-adapter-portfolio.html

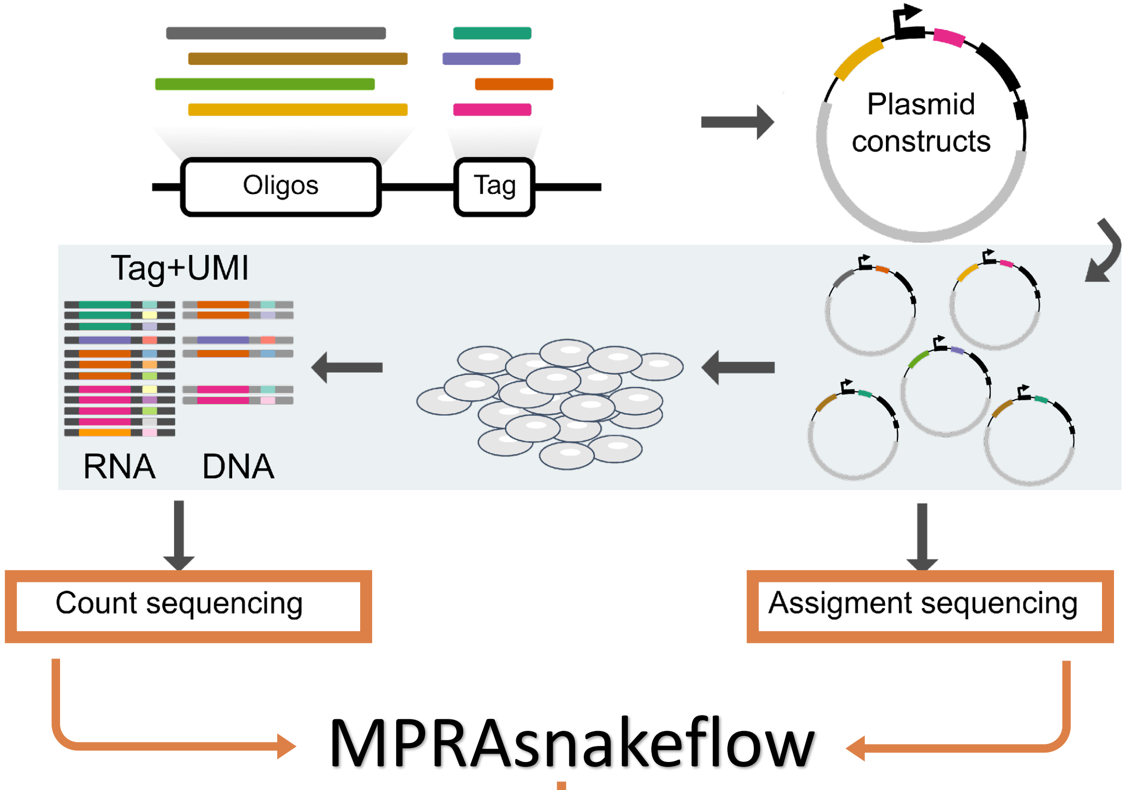
MPRAsnakeflow – Pipeline to process sequencing data from MPRA to create count tables for candidate sequences and variants



Getting a count matrix

ID	Barcode	DNA1	DNA2	DNA3	DNA4	DNA5	RNA1	RNA2	RNA3	RNA4	RNA5
Oligo1	ACTCGAAACCAGGTCCCATT	9	9	5	16	6	6	7	13	8	18
Oligo1	ACTGGACAATTTAATAGCAT	1	0	0	0	1	0	0	0	0	3
Oligo1	ACTTTATGCAACACGTAAAG	1	1	0	0	1	1	1	0	0	1
Oligo1	AGAACATATCTGGCACATGA	23	14	18	5	13	11	36	3	35	32
Oligo1	AGACGGCTTACGAGGCATCC	16	8	15	13	11	20	29	14	15	19
Oligo1	AGCAACAACAGGTCAACCAG	24	24	22	15	18	2610	3830	4760	1215	1935
Oligo1	AGGGGGAGTACCGAGTGAAT	2	7	0	3	2	1	7	2	5	7
Oligo1	AGTAAACACCGCATTATGGC	8	5	3	7	1	2	7	5	8	3
Oligo1	AGTAGCGTAAACCTCCCCCG	9	6	9	7	11	9	23	3	16	15
Oligo1	AGTCCATCTGAATACACGTA	21	13	22	14	26	14	50	31	33	31
Oligo1	ATACCTTACTAACACAACTT	7	3	3	5	7	2	15	3	4	5
Oligo1	ATATATCAAACGCGGATACA	2	6	4	2	7	5	12	6	12	12
Oligo1	ATCAGTACATTGTCACATAT	6	4	2	1	5	6	8	4	15	12
Oligo1	ATCGTCTTCCCGCAAAGGCA	4	2	7	5	1	1	3	3	6	8
Oligo1	ATCGTTAGGGCAAGGAGCAC	37	21	40	23	38	31	116	67	53	64

Statistical analysis of counts and identifying significant variant and element effects

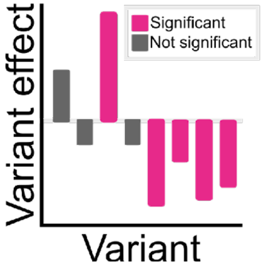


$$\mathrm{DNA} \sim Gamma(k, b)$$

$$\mathrm{RNA}|\mathrm{DNA} \sim Poisson(\alpha \cdot \mathrm{DNA})$$

$$\mathrm{RNA} \sim NB\left(\mu = \frac{\alpha \cdot k}{\beta}, \psi = k\right)$$

BCalm, mpralm, MPRAnalyze, ...



Thank You! Questions?

kircherlab.github.io

CONTACT

Regulatory Genomics & Computational Genome Biology

Martin Kircher martin.kircher
@uni-luebeck.de | @uksh.de | @bih-charite.de

UNIVERSITÄT ZU LÜBECK

UK SH UNIVERSITÄTSKLINIKUM Schleswig-Holstein

BIH Berlin Institute of Health @Charité


COFFEE
BREAK